

Inverse & Composite Functions

ANSWER KEY



Composing functions

- 1 Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$$(f \circ g)(x) = f(g(x)) = 2(x^2 - 1) + 3 = 2x^2 + 1.$$

$$(g \circ f)(x) = g(f(x)) = (2x + 3)^2 - 1 = 4x^2 + 12x + 9 - 1 = 4x^2 + 12x + 8.$$

- 2 Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(2)$ and state the domain restriction that makes $f \circ g$ defined.

$(f \circ g)(2) = f(g(2)) = f(0) = \sqrt{0 + 4} = 2$. In general, $f \circ g$ requires $g(x) + 4 \geq 0$, i.e. $x^2 \geq 0$, which holds for all real x — so $f \circ g$ is defined everywhere.

Decomposing a composite function

- 3 Write $h(x) = (3x - 5)^4$ as a composition $h(x) = f(g(x))$ for two simpler functions f and g .

Let $g(x) = 3x - 5$ (inner function) and $f(x) = x^4$ (outer function). Then $f(g(x)) = (3x - 5)^4 = h(x)$.

- 4 Write $h(x) = \sqrt{x^2 + 1}$ as a composition of two functions.

Let $g(x) = x^2 + 1$ and $f(x) = \sqrt{x}$. Then $f(g(x)) = \sqrt{x^2 + 1} = h(x)$.

Verifying inverse functions

- 5 Verify algebraically that $f(x) = \frac{x - 2}{3}$ and $g(x) = 3x + 2$ are inverses of each other.

$f(g(x)) = \frac{(3x + 2) - 2}{3} = \frac{3x}{3} = x$. $g(f(x)) = 3\left(\frac{x - 2}{3}\right) + 2 = (x - 2) + 2 = x$. Since both compositions return x , f and g are inverses.

- 6 Find the inverse of $f(x) = 2x^3 + 1$, and verify that $f(f^{-1}(x)) = x$ for one value, say $x = 17$.

Solve $y = 2x^3 + 1$ for x : $x^3 = \frac{y - 1}{2}$, so $f^{-1}(x) = \sqrt[3]{\frac{x - 1}{2}}$. Check at $x = 17$: $f^{-1}(17) = \sqrt[3]{\frac{17 - 1}{2}} = \sqrt[3]{8} = 2$, and $f(2) = 2(8) + 1 = 17$ ✓.

Visualizing inverse functions as reflections

- 7 The graph shows $f(x) = x^2$ (restricted to $x \geq 0$) and its inverse $f^{-1}(x) = \sqrt{x}$, which are reflections of each other across the line $y = x$.
- a** State the domain restriction placed on $f(x) = x^2$ needed for it to have an inverse, and explain why. $f(x) = x^2$ must be restricted to $x \geq 0$ (or $x \leq 0$) because the full parabola fails the horizontal line test — without restricting the domain, f is not one-to-one and has no inverse function.
- b** Confirm from the graph that f and f^{-1} intersect on the line $y = x$, and find that intersection point algebraically. Setting $f(x) = x$: $x^2 = x$, so $x^2 - x = 0$, $x(x - 1) = 0$, giving $x = 0$ or $x = 1$. Both points $(0, 0)$ and $(1, 1)$ lie on $y = x$, matching where the two curves cross in the graph.

Domain and range of composite and inverse functions

- 8 Given $f(x) = \frac{1}{x-3}$ and $g(x) = x + 1$, find the domain of $(f \circ g)(x)$.
 $(f \circ g)(x) = f(g(x)) = \frac{1}{(x+1)-3} = \frac{1}{x-2}$. This is undefined at $x = 2$, so the domain is all reals except $x = 2$.
- 9 If f has domain $[1, \infty)$ and range $[0, \infty)$, state the domain and range of f^{-1} .
 The domain and range swap for an inverse function: f^{-1} has domain $[0, \infty)$ and range $[1, \infty)$.

Applying composite and inverse functions in context

- 10 A temperature-conversion function $C(F) = \frac{5}{9}(F - 32)$ converts Fahrenheit to Celsius. Find $C^{-1}(F)$ and interpret what it computes.
 Solve $C = \frac{5}{9}(F - 32)$ for F : $\frac{9C}{5} = F - 32$, so $F = \frac{9}{5}C + 32$. This is C^{-1} , which converts a Celsius temperature back to Fahrenheit.

Using the horizontal line test

- 11 Determine whether $f(x) = x^3 - 4x$ has an inverse function over its entire domain. Justify using the horizontal line test.
 Since $f(x) = x^3 - 4x$ has multiple turning points (a cubic with two real critical points), a horizontal line such as $y = 0$ intersects the graph at more than one point ($x = 0, \pm 2$), failing the horizontal line test. So f does not have an inverse function over its entire domain.
- 12 Explain why $f(x) = 5x - 2$ passes the horizontal line test for all real x , and find its inverse.
 $f(x) = 5x - 2$ is a line with nonzero slope, so it is strictly increasing and every horizontal line crosses it exactly once — it passes the horizontal line test everywhere. Solving $y = 5x - 2$ for x : $f^{-1}(x) = \frac{x+2}{5}$.

Composite functions with restricted domains

- 13 Given $f(x) = \frac{1}{x}$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and state its domain.

$(f \circ g)(x) = f(x - 4) = \frac{1}{x - 4}$. This is undefined at $x = 4$, so the domain is all reals except $x = 4$.

Multiple choice — easy 15

- 14 Given $f(x) = x - 1$ and $g(x) = 5x - 4$, find $(f \circ g)(-6)$.

- a. -34
- b. -36
- c. -35 ✓
- d. -39

$$(f \circ g)(-6) = f(g(-6)) = f(-34) = 1(-34) - 1 = -35.$$

- 15 Given $f(x) = -3x - 6$ and $g(x) = 6x$, find $(f \circ g)(-4)$.

- a. 67
- b. 65
- c. 36
- d. 66 ✓

$$(f \circ g)(-4) = f(g(-4)) = f(-24) = -3(-24) - 6 = 66.$$

- 16 Given $f(x) = 5x - 2$ and $g(x) = x + 2$, find $(f \circ g)(-2)$.

- a. -3
- b. -10
- c. -2 ✓
- d. -1

$$(f \circ g)(-2) = f(g(-2)) = f(0) = 5(0) - 2 = -2.$$

17 Given $f(x) = 9x - 3$ and $g(x) = 8x + 1$, find $(f \circ g)(2)$.

a. 151

b. 149

c. 121

d. 150 ✓

$$(f \circ g)(2) = f(g(2)) = f(17) = 9(17) - 3 = 150.$$

18 Given $f(x) = 9x + 5$ and $g(x) = -8x - 9$, find $(f \circ g)(-2)$.

a. 69

b. 67

c. 68 ✓

d. 95

$$(f \circ g)(-2) = f(g(-2)) = f(7) = 9(7) + 5 = 68.$$

19 Given $f(x) = 9x - 5$ and $g(x) = -9x - 6$, find $(f \circ g)(-1)$.

a. 21

b. 22 ✓

c. 23

d. 120

$$(f \circ g)(-1) = f(g(-1)) = f(3) = 9(3) - 5 = 22.$$

20 Given $f(x) = -9x + 2$ and $g(x) = -8x - 1$, find $(f \circ g)(6)$.

a. 442

b. 444

c. 415

d. 443 ✓

$$(f \circ g)(6) = f(g(6)) = f(-49) = -9(-49) + 2 = 443.$$

21 Given $f(x) = -8x - 6$ and $g(x) = 6x - 7$, find $(f \circ g)(6)$.

a. -239

b. -331

c. -238 ✓

d. -237

$$(f \circ g)(6) = f(g(6)) = f(29) = -8(29) - 6 = -238.$$

22 Given $f(x) = 4x - 1$ and $g(x) = -7x - 3$, find $(f \circ g)(0)$.

a. -13 ✓

b. -14

c. 4

d. -12

$$(f \circ g)(0) = f(g(0)) = f(-3) = 4(-3) - 1 = -13.$$

23 Given $f(x) = -3x - 2$ and $g(x) = -5x + 1$, find $(f \circ g)(3)$.

a. 56

b. 39

c. 40 ✓

d. 41

$$(f \circ g)(3) = f(g(3)) = f(-14) = -3(-14) - 2 = 40.$$

24 Given $f(x) = 6x - 9$ and $g(x) = -x - 1$, find $(f \circ g)(5)$.

a. -46

b. -22

c. -45 ✓

d. -44

$$(f \circ g)(5) = f(g(5)) = f(-6) = 6(-6) - 9 = -45.$$

25 Given $f(x) = 5x - 6$ and $g(x) = x + 8$, find $(f \circ g)(1)$.

a. 7

b. 39 ✓

c. 40

d. 38

$$(f \circ g)(1) = f(g(1)) = f(9) = 5(9) - 6 = 39.$$

26 Given $f(x) = 6x + 4$ and $g(x) = 9x - 6$, find $(f \circ g)(1)$.

a. 84

b. 21

c. 23

d. 22 ✓

$$(f \circ g)(1) = f(g(1)) = f(3) = 6(3) + 4 = 22.$$

27 Given $f(x) = -5x - 4$ and $g(x) = 6x - 1$, find $(f \circ g)(6)$.

a. -179 ✓

b. -178

c. -205

d. -180

$$(f \circ g)(6) = f(g(6)) = f(35) = -5(35) - 4 = -179.$$

28 Given $f(x) = -3x - 8$ and $g(x) = 4x - 5$, find $(f \circ g)(1)$.

- a. -4
- b. -6
- c. -5 ✓
- d. -49

$$(f \circ g)(1) = f(g(1)) = f(-1) = -3(-1) - 8 = -5.$$

Multiple choice — medium 15

29 Given $f(x) = -2x - 9$, find $f^{-1}(-5)$.

- a. -2 ✓
- b. -3
- c. -1
- d. -5

$$\text{Solve } -5 = -2x - 9 \text{ for } x: x = \frac{-5 + 9}{-2} = -2, \text{ so } f^{-1}(-5) = -2.$$

30 Given $f(x) = 9x - 1$, find $f^{-1}(-73)$.

- a. -8 ✓
- b. -73
- c. -7
- d. -9

$$\text{Solve } -73 = 9x - 1 \text{ for } x: x = \frac{-73 + 1}{9} = -8, \text{ so } f^{-1}(-73) = -8.$$

31 Given $f(x) = 6x - 9$, find $f^{-1}(39)$.

a. 8 ✓

b. 9

c. 7

d. 39

Solve $39 = 6x - 9$ for x : $x = \frac{39 + 9}{6} = 8$, so $f^{-1}(39) = 8$.

32 Given $f(x) = -4x + 5$, find $f^{-1}(29)$.

a. -6 ✓

b. -7

c. -5

d. 29

Solve $29 = -4x + 5$ for x : $x = \frac{29 - 5}{-4} = -6$, so $f^{-1}(29) = -6$.

33 Given $f(x) = -5x + 4$, find $f^{-1}(44)$.

a. 44

b. -7

c. -8 ✓

d. -9

Solve $44 = -5x + 4$ for x : $x = \frac{44 - 4}{-5} = -8$, so $f^{-1}(44) = -8$.

34 Given $f(x) = 2x + 6$, find $f^{-1}(-4)$.

a. -6

b. -5 ✓

c. -4

d. -7

Solve $-4 = 2x + 6$ for x : $x = \frac{-4 - 6}{2} = -5$, so $f^{-1}(-4) = -5$.

35 Given $f(x) = -3x + 8$, find $f^{-1}(8)$.

a. 8

b. 0 ✓

c. -1

d. 1

Solve $8 = -3x + 8$ for x : $x = \frac{8 - 8}{-3} = 0$, so $f^{-1}(8) = 0$.

36 Given $f(x) = -8x + 7$, find $f^{-1}(-25)$.

a. 4 ✓

b. 3

c. -25

d. 5

Solve $-25 = -8x + 7$ for x : $x = \frac{-25 - 7}{-8} = 4$, so $f^{-1}(-25) = 4$.

37 Given $f(x) = -9x - 5$, find $f^{-1}(-68)$.

a. 8

b. 7 ✓

c. 6

d. -68

Solve $-68 = -9x - 5$ for x : $x = \frac{-68 + 5}{-9} = 7$, so $f^{-1}(-68) = 7$.

38 Given $f(x) = 6x + 7$, find $f^{-1}(-11)$.

- a. -11
- b. -3 ✓
- c. -4
- d. -2

Solve $-11 = 6x + 7$ for x : $x = \frac{-11 - 7}{6} = -3$, so $f^{-1}(-11) = -3$.

39 Given $f(x) = 7x + 7$, find $f^{-1}(-21)$.

- a. -21
- b. -3
- c. -5
- d. -4 ✓

Solve $-21 = 7x + 7$ for x : $x = \frac{-21 - 7}{7} = -4$, so $f^{-1}(-21) = -4$.

40 Given $f(x) = -5x + 8$, find $f^{-1}(23)$.

- a. -3 ✓
- b. -2
- c. 23
- d. -4

Solve $23 = -5x + 8$ for x : $x = \frac{23 - 8}{-5} = -3$, so $f^{-1}(23) = -3$.

41 Given $f(x) = 2x - 9$, find $f^{-1}(-19)$.

- a. -5 ✓
- b. -4
- c. -19
- d. -6

Solve $-19 = 2x - 9$ for x : $x = \frac{-19 + 9}{2} = -5$, so $f^{-1}(-19) = -5$.

42 Given $f(x) = 8x - 3$, find $f^{-1}(-3)$.

a. -1

b. 1

c. -3

d. 0 ✓

Solve $-3 = 8x - 3$ for x : $x = \frac{-3 + 3}{8} = 0$, so $f^{-1}(-3) = 0$.

43 Given $f(x) = -9x + 4$, find $f^{-1}(4)$.

a. -1

b. 1

c. 4

d. 0 ✓

Solve $4 = -9x + 4$ for x : $x = \frac{4 - 4}{-9} = 0$, so $f^{-1}(4) = 0$.

Multiple choice — hard 15

44 Given $f(x) = 5x - 7$ and $g(x) = -x^2 + 5$, find $(g \circ f)(-3)$.

a. -477

b. 27

c. -481

d. -479 ✓

$(g \circ f)(-3) = g(f(-3)) = g(-22) = -1(-22)^2 + 5 = -479$.

45 Given $f(x) = -5x - 8$ and $g(x) = 5x^2 - 6$, find $(g \circ f)(-1)$.

- a. 41
- b. -21
- c. 37
- d. 39 ✓

$$(g \circ f)(-1) = g(f(-1)) = g(-3) = 5(-3)^2 - 6 = 39.$$

46 Given $f(x) = -4x + 4$ and $g(x) = -x^2 + 7$, find $(g \circ f)(-3)$.

- a. -247
- b. -249 ✓
- c. -9
- d. -251

$$(g \circ f)(-3) = g(f(-3)) = g(16) = -1(16)^2 + 7 = -249.$$

47 Given $f(x) = -2x - 1$ and $g(x) = 5x^2$, find $(g \circ f)(1)$.

- a. 45 ✓
- b. 43
- c. -15
- d. 47

$$(g \circ f)(1) = g(f(1)) = g(-3) = 5(-3)^2 = 45.$$

48 Given $f(x) = -3x - 8$ and $g(x) = 3x^2 - 4$, find $(g \circ f)(0)$.

- a. 190
- b. 188 ✓
- c. -28
- d. 186

$$(g \circ f)(0) = g(f(0)) = g(-8) = 3(-8)^2 - 4 = 188.$$

49 Given $f(x) = -7x$, find the slope and y -intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $(-7, 0)$

b. $(-\frac{1}{7}, 0)$ ✓

c. $(\frac{1}{7}, 0)$

d. $(\frac{5}{14}, 0)$

Solve $y = -7x$ for x : $x = \frac{y}{-7}$. So $f^{-1}(x)$ has slope $\frac{1}{-7} = -\frac{1}{7}$ and intercept 0: $(-\frac{1}{7}, 0)$.

50 Given $f(x) = 5x - 5$, find the slope and y -intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $(\frac{1}{5}, -1)$

b. $(\frac{1}{5}, 1)$ ✓

c. $(5, -5)$

d. $(-\frac{1}{5}, 1)$

Solve $y = 5x - 5$ for x : $x = \frac{y+5}{5}$. So $f^{-1}(x)$ has slope $\frac{1}{5} = \frac{1}{5}$ and intercept 1: $(\frac{1}{5}, 1)$.

51 Given $f(x) = -9x - 1$, find the slope and y -intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $(\frac{1}{9}, -\frac{1}{9})$

b. $(-9, -1)$

c. $(-\frac{1}{9}, -\frac{1}{9})$ ✓

d. $(-\frac{1}{9}, \frac{1}{9})$

Solve $y = -9x - 1$ for x : $x = \frac{y+1}{-9}$. So $f^{-1}(x)$ has slope $\frac{1}{-9} = -\frac{1}{9}$ and intercept $-\frac{1}{9}$: $(-\frac{1}{9}, -\frac{1}{9})$.

52 Given $f(x) = 2x + 4$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $(\frac{1}{2}, -2)$ ✓

b. $(-\frac{1}{2}, -2)$

c. $(2, 4)$

d. $(\frac{1}{2}, 2)$

Solve $y = 2x + 4$ for x : $x = \frac{y-4}{2}$. So $f^{-1}(x)$ has slope $\frac{1}{2} = \frac{1}{2}$ and intercept -2 : $(\frac{1}{2}, -2)$.

53 Given $f(x) = 6x - 5$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $(-\frac{1}{6}, \frac{5}{6})$

b. $(6, -5)$

c. $(\frac{1}{6}, -\frac{5}{6})$

d. $(\frac{1}{6}, \frac{5}{6})$ ✓

Solve $y = 6x - 5$ for x : $x = \frac{y+5}{6}$. So $f^{-1}(x)$ has slope $\frac{1}{6} = \frac{1}{6}$ and intercept $\frac{5}{6}$: $(\frac{1}{6}, \frac{5}{6})$.

54 Determine whether $f(x) = x^3 - 6x$ has an inverse function over its entire domain, using the horizontal line test.

a. Has an inverse function over its entire domain (strictly increasing)

b. **Does NOT have an inverse function over its entire domain (not one-to-one)** ✓

c. Has an inverse only on the interval $x \geq 0$

d. Cannot be determined without a graph

Since $b = -6 < 0$, $f(x) = x^3 - 6x$ has two turning points (critical points where $3x^2 - 6 = 0$), so some horizontal lines cross the graph more than once. Does NOT have an inverse function over its entire domain (not one-to-one).

- 55 Determine whether $f(x) = x^3 - 3x$ has an inverse function over its entire domain, using the horizontal line test.

- a. **Does NOT have an inverse function over its entire domain (not one-to-one) ✓**
- b. Has an inverse only on the interval $x \geq 0$
- c. Cannot be determined without a graph
- d. Has an inverse function over its entire domain (strictly increasing)

Since $b = -3 < 0$, $f(x) = x^3 - 3x$ has two turning points (critical points where $3x^2 - 3 = 0$), so some horizontal lines cross the graph more than once. Does NOT have an inverse function over its entire domain (not one-to-one).

- 56 Determine whether $f(x) = x^3 + x$ has an inverse function over its entire domain, using the horizontal line test.

- a. Has an inverse only on the interval $x \geq 0$
- b. **Has an inverse function over its entire domain (strictly increasing) ✓**
- c. Does NOT have an inverse function over its entire domain (not one-to-one)
- d. Cannot be determined without a graph

Since $b = 1 > 0$, $f(x) = x^3 + x$ is strictly increasing everywhere (the cubic term dominates and there are no turning points), so every horizontal line crosses it exactly once. Has an inverse function over its entire domain (strictly increasing).

- 57 Determine whether $f(x) = x^3 - 2x$ has an inverse function over its entire domain, using the horizontal line test.

- a. **Does NOT have an inverse function over its entire domain (not one-to-one) ✓**
- b. Cannot be determined without a graph
- c. Has an inverse function over its entire domain (strictly increasing)
- d. Has an inverse only on the interval $x \geq 0$

Since $b = -2 < 0$, $f(x) = x^3 - 2x$ has two turning points (critical points where $3x^2 - 2 = 0$), so some horizontal lines cross the graph more than once. Does NOT have an inverse function over its entire domain (not one-to-one).

- 58 Determine whether $f(x) = x^3 + 2x$ has an inverse function over its entire domain, using the horizontal line test.
- a. Has an inverse only on the interval $x \geq 0$
 - b. Does NOT have an inverse function over its entire domain (not one-to-one)
 - c. Cannot be determined without a graph
 - d. **Has an inverse function over its entire domain (strictly increasing) ✓**

Since $b = 2 > 0$, $f(x) = x^3 + 2x$ is strictly increasing everywhere (the cubic term dominates and there are no turning points), so every horizontal line crosses it exactly once. Has an inverse function over its entire domain (strictly increasing).